

User Capacity & Concurrent Traffic Analysis

Platform Architecture: Vercel Serverless & Edge Network

Target Application: Plexuspharmaco Corporate Portal

1. Estimated Concurrent User Capacity

Edge CDN / Static Assets: **Practically Unlimited (10,000+ Concurrent Users)**

Because the site is built with Next.js and deployed on Vercel, static pages, images (via `next/image`), and preloaded fonts are served directly from global Edge nodes caching the content worldwide.

Serverless Functions / API Routes: **1,000+ Concurrent Executions (Auto-scaling)**

Vercel automatically scales serverless functions up or down depending on incoming traffic spikes. Standard hobby/pro limits allow rapid scaling to handle heavy concurrent hits without crashing.

2. Infrastructure Capacity Breakdown

Traffic Type	Estimated Concurrent Limit	Bottleneck Factor
Page Views (Static/Cached)	10,000+ concurrent visitors	User's local network bandwidth / CDN edge limits.
Dynamic Forms / API Calls	1,000+ concurrent requests	Backend integration or email delivery service limits (e.g., Contact/Business Enquiry endpoints).
Media / Asset Delivery	Unlimited (Global CDN)	Edge caching efficiency.

3. Recommendations to Handle High Traffic Surges

- **Enable Incremental Static Regeneration (ISR):** Keeps pages lightning fast even during massive traffic spikes by serving cached versions while updating in the background.
- **Rate Limiting on Forms:** Protect "Business Enquiry" and contact endpoints against automated bot spikes or denial-of-service attempts.
- **Monitor Vercel Analytics:** Track real-time visitor concurrency and serverless function execution durations under peak loads.